温州肯恩大学
WENZHOU-KEAN UNIVERSITY

MGS 3101 W03 Foundations of Business Analytics

Data Wrangling I

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Week 7-1, 14 Oct 2025

Outline

1. Opening / Hook
   → Engage with messy data examples
2. What is Data Wrangling
   → Definition, scope
3. Why Data Wrangling Matters
   → Importance, benefits, and real-world relevance
4. How Data Wrangling Works
   → Six steps: discovering, structuring, cleaning, enriching, verifying, publishing
5. Focus: Data Cleaning
   → Missing values, normalization & scaling, outliners

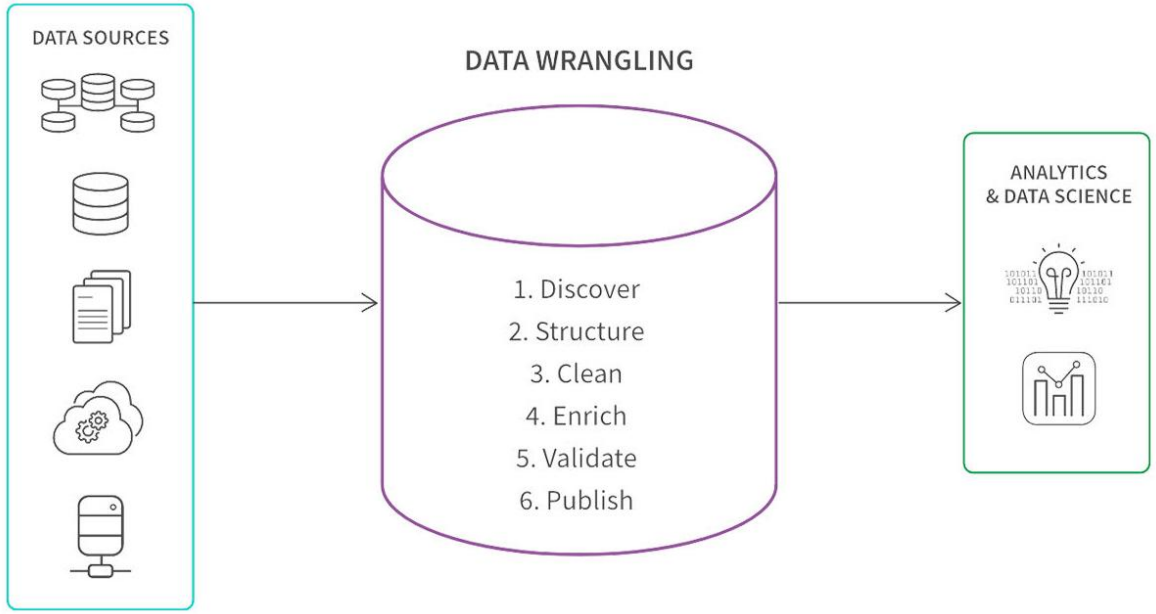
Question: What **issues** or **inconsistencies** do you see in this dataset?

Name	Gender	Age	Join_Date	Purchase_Amount	Country
Alice Wang	Female	25	2023-05-12	320.5	China
bob chen	M	22	05/13/2023	28	CHN
Charlie Zhang	male	-19	2023/14/05		China
David Lee	NaN	27	2023-05-15	9999	USA
Alice Wang	Female	25	2023-05-12	320.5	China
Eva Liu	Female	230	2023-05-20	400	Singapore
Frank O'Neil	Male	29	2023-05-19	N/A	UK
李雷	male	30	2023-05-21	350.0	China
Kim, Grace	FEMALE		2023-05-25	270	Korea
Henry Chan	Male	28	2023-05-12	310	China

- Inconsistent data format/capitalization/country codes
- Missing values and/or non-numeric value
- Unrealistic values and/or outliners
- Encoding/non-English name
- Duplicates

What Is Data Wrangling?

- Definition
  - The process of **cleaning**, **transforming**, and **organizing** raw data into accurate, reliable and analysis-ready formats for data science and analytics
  - Also known as: *data cleaning*, *data remediation*, *data munging*
- Raw Data
  - Unprocessed or unprepared for analysis
  - Often messy, inconsistent, or incomplete
  - Also known as: *source data*, *primary data*



Source: https://www.qlik.com/us/data-management/data-wrangling

Why Data Wrangling Is Important?

✓ Better Data Quality

- Fixes missing values, outliers, and inconsistencies
- Standardizes formats, units, and categories for reliable analysis

⚙ Higher Efficiency

- Less time cleaning, more time analyzing and exploring pattern

🔗 Easier Integration

- Combines data from multiple sources and formats seamlessly

📈 Ready for Data Analysis

- Provide clean, structured inputs for accurate model training

💡 Informed Decisions

- High quality data → trustworthy insights → better business outcomes

Source: https://www.qlik.com/us/data-management/data-wrangling

Six Steps of Data Wrangling

1) Discovering
   - Explore and **understand** raw data – its structure, content, and potential use
2) Structuring
   - **Arrange** raw data into a usable format for analysis
3) Cleaning
   - **Detect** and **correct** errors, inconsistences, missing values, duplicates, etc.
4) Enriching
   - **Add** new information or **transform** existing data to enhance value
5) Validating
   - **Check** accuracy, reliability, and readiness for analysis
6) Publishing
   - **Save** and **share** the clean, well-documented dataset for use

Data Wrangling – Discovering

🎯 Goal

- Understand the raw data — its **sources**, **structure**, and potential **use**

📝 Key Questions

- Where does the data come from? Databases (Access, Sql, Oracle), APIs, web scraping, web pages/html files, txt/csv/json/xml/xlsx files
- What formats and structures does it have?
- How will it be used in analysis?

- Focus
  - Identify quality issues such as *missing values*, *inconsistencies*, *errors*, *biases*, or *outliers* that may affect analysis
- Outcome
  - A **data quality or profiling report** summarizing key statistics, distributions, and findings

Data Wrangling – Discovering

- Common File Types
  - Delimited text: `.txt` , `.csv`
  - Structured files: `.xlsx` (Excel), databases (Access, SQL, Oracle)
  - Semi-structured files: `.json` , `.xml`, web/HTML pages
  - Other sources: APIs, web scraping, cloud storage
- Data Input in Pandas

```
pd.read_csv()       # CSV or text files
pd.read_excel()     # Excel files
pd.read_sql()       # Databases
pd.read_json()      # JSON files
```

Data Formats

- **Structured**: organized in tables (rows & columns) → easy to query
- **Unstructured**: free-form (text, images, video) → hard to parse directly
- **Semi-structured**: flexible format with tags or attributes (e.g., JSON, XML)

Data Wrangling – Structuring

🎯 Goal

- Organize and transform data into a *consistent*, *analysis-ready format*

📝 Key Tasks

- **Aggregation:** summarize data by groups
  → `df.groupby()`, `df.agg()`, `df.mean()`, `df.sum()`
- **Pivoting:** reshape data between rows and columns
  → `df.pivot()`, `df.pivot_table()`, `df.melt()`
- **Joining:** merge data from multiple tables or sources
  → `df.merge()`, `df.join()`, `df.concat()`
- **Data Type Conversion:** ensure correct data types for computation
  → `df.astype()`, `df.to_datetime()`, `df.to_numeric()`

Data Wrangling – Structuring

Key Pandas Functions:

- `pd.melt()`: stack (wide → long)
- `df.pivot()` or `pd.pivot_table()`: unstack (long → wide)

Stacked (Long) Data

- Each variable's values are stored in one single column
- Each row represents one observation of a variable
- Commonly used for *statistical analysis and visualization*

	A	B	C
1	**Team**	**Year**	**Three Point Attempts**
2	Gamaliel Gremlins	1	602
3	Fitzgerald Hawks	1	713
4	Delano Demons	1	647
5	Ulysses Gazelles	1	711
6	Milhous Rams	1	800
7	Birchard Spiders	1	682
8	Quincy Bobcats	1	800
9	Knox Golden Bears	1	650
10	Gamaliel Gremlins	2	918

Unstacked (Wide) Data

- Each variable's values appear in separate columns
- Easier for *human reading or reporting*

	A	B	C	D	E	F	G
1		**Three Point Attempts**					
2	**Team**	**Year 1**	**Year 2**	**Year 3**	**Year 4**	**Year 5**	**Year 5**
3	Gamaliel Gremlins	602	918	776	590	600	559
4	Fitzgerald Hawks	713	755	808	841	831	628
5	Delano Demons	647	759	653	779	626	826
6	Ulysses Gazelles	711	873	919	676	775	645
7	Milhous Rams	800	890	506	695	775	764
8	Birchard Spiders	682	762	587	800	903	773
9	Quincy Bobcats	800	631	626	910	894	670
10	Knox Golden Bears	650	794	737	772	642	717

Data Wrangling – Cleaning

We will perform **data cleaning using Python** in the next class.

Goal

- Ensure data are *accurate*, *consistent*, and *reliable* before analysis

Key Tasks

- Handle **missing values**: detect and decide how to treat
- Remove **duplicates**: find and drop
- **Data type** conversions: e.g., convert to datetime
- **Text cleaning**: trim spaces, unify case, remove symbols, use regex
- **Categorical** processing: create dummy or ordinal variables
- **Normalization & scaling**: apply min-max, z-score, or robust scaling (median, IQR)
- Handle **outliners**: detect and decide to keep or remove (z-score, IQR)
- Data **transformation**: log, square root, power, binning, etc.

- Tips
  - Identifying and removing any bad data greatly impacts the rest of the wrangling processes.
  - Avoid overcleaning — do not remove useful variation or rare patterns

Data Cleaning – Missing Values

How would you handle missing values or incomplete observations?

Name	Gender	Age	Join_Date	Purchase_Amount	Country
Alice Wang	Female	25	2023-05-12	320.5	China
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Data Cleaning – Missing Values

Three Types of Missing Mechanisms (Rubin 1976)

1. Missing Completely at Random (MCAR)

- Definition
  - Missingness is **unrelated to any variable**, neither observed nor unobserved.
- Example
  - A few survey responses are lost due to a system glitch – every participant has an equal chance of being affected.
- Notes
  - This is a strong and often unrealistic assumption – real-world missingness is rarely completely random.
  - If data are truly MCAR, deleting missing observations (listwise deletion) will **not bias** the analysis results.

Data Cleaning – Missing Values

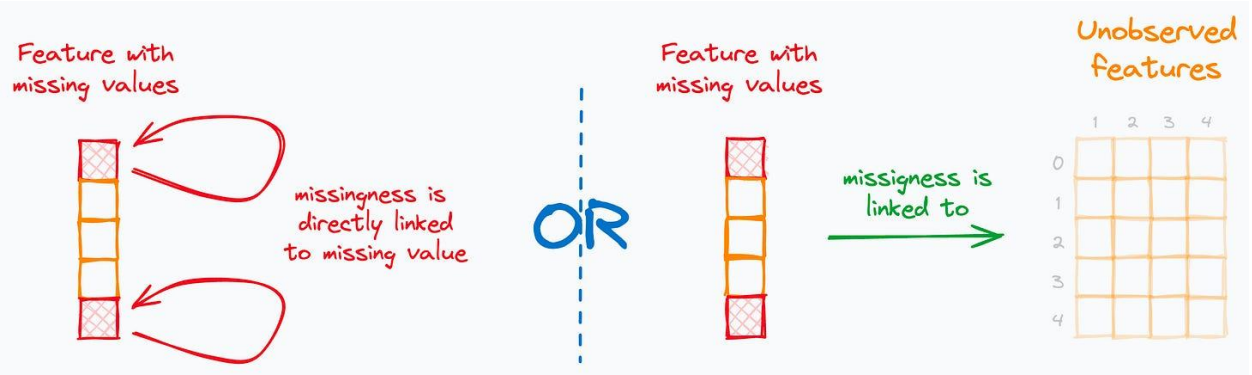
https://blog.dailydoseofds.com/p/3-types-of-missing-values

2. Missing at Random (MAR)

- Definition
  - Missingness is **related to other observed variables**, but **not to the missing value itself**.
- Example (missing in income)
  - In a survey, older respondents are less likely to report income, but within each age group, missingness occurs randomly.
- Notes
  - MAR is common and realistic in many real-world datasets.
  - Missing values can be handled using imputation methods, such as mean, regression, KNN, MICE, Miss Forest, etc.

- Extensive research exists on imputation methods.
- In practice, analysts commonly use appropriate imputation techniques to address missing values.

Data Cleaning – Missing Values



https://blog.dailydoseofds.com/p/3-types-of-missing-values

3. Missing Not at Random (MNAR)

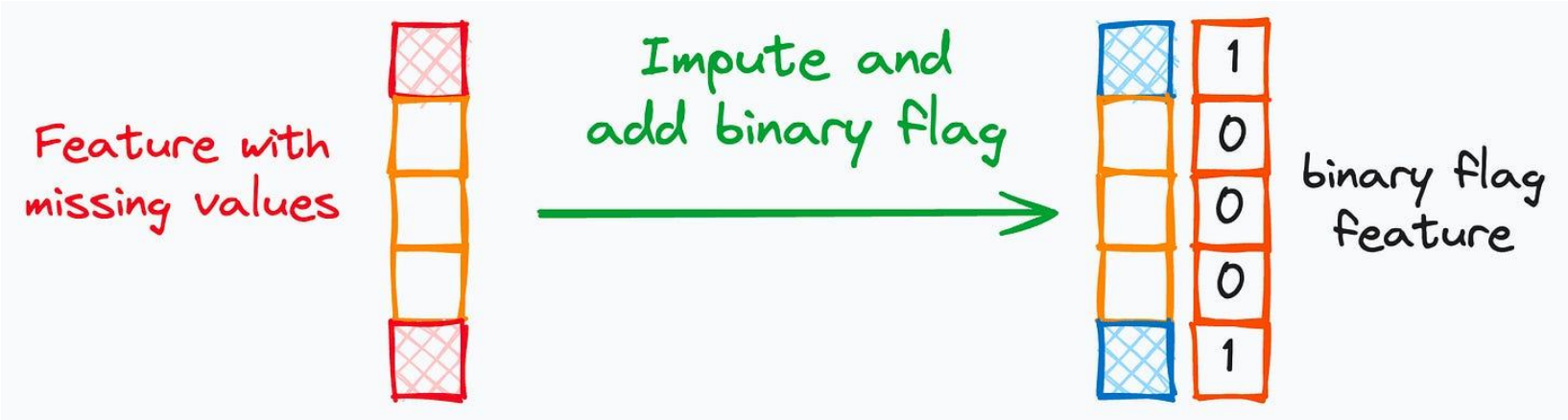
- Definition
  - Missingness is **related to the value that is missing itself**, or **the unobserved variables**.
- Example (missing in income)
  - In a survey, people with very high income may choose not to disclose their income.
- Notes
  - MNAR is the hardest to handle – bias remains even after imputation.
  - Require modeling the missing mechanism or collecting additional data to adjust properly.

Data Cleaning – Missing Values

A Useful Practice

- In **MNAR**, the **pattern of missingness** itself may contain valuable information.
- Direct imputation can remove this valuable signal.

✅ To retain it, **add a binary indicator variable** (e.g., `is_missing = 1` if value was imputed, else `0`) to preserve information about missingness. Include this dummy variable in your model analysis to assess how missingness affects the dependent variable or related to your research objective.



https://blog.dailydoseofds.com/p/3-types-of-missing-values

Data Cleaning – Missing Values

It is essential for data scientists to collaborate with data engineers and understand the **data collection process**.

https://blog.dailydoseofds.com/p/3-types-of-missing-values

Data Cleaning – Normalization and Scaling

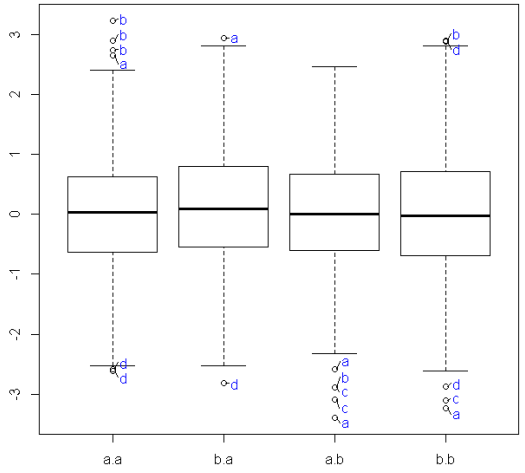
🎯 Goal

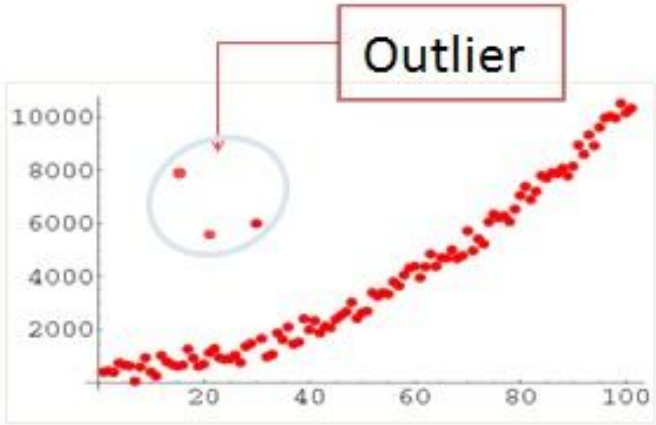
- Ensure that numerical features are on a comparable scale, improving model **performance** and **interpretability**

- Normalization
  - Transform features to a fixed range, typically **[0,1]**
  - Common methods: *Min-Max, Max-Abs, Mean, Z-score, Log*
  - Useful when features have **different units** or **magnitudes**
- Scaling (Broader Concept)
  - Encompasses both normalization and standardization
  - Adjusts the **spread** or **variability** of data rather than its range
  - Why scale?
    - **Robustness to Outliers:** reduce sensitivity to extreme values
    - **Algorithm Compatibility:** Some algorithms, like Support Vector Machines (SVM) and Principal Component Analysis (PCA), work best with scaled data

Source: https://www.geeksforgeeks.org/data-analysis/normalization-and-scaling/

Data Cleaning – Outliners





- Definition
  - **Unusually small** or **large** values
  - No single universal definition, but statistical rules help us identify them
- Identification (Rules of Thumb)
  - z-scores rule: If $|\boldsymbol{z}| > \mathbf{3}$, the value may be an outliner
  - Interquartile Range (IQR) rule: $IQR = Q_3 - Q_1$
    - Mild outliners: $< Q_1 - 1.5 * IQR$ or $> Q_3 + 1.5 * IQR$
    - Extreme outliner: $< Q_1 - 3 * IQR$ or $> Q_3 + 3 * IQR$
  - Boxplot
- Possible Causes
  - Data entry error
  - Incorrectly recorded or included observation
  - Genuine extreme value (important to the analysis)
- Handling Outliners
  - **Keep** them if they represent valid but unusual cases
  - **Remove** them if they are clear mistakes or irrelevant
  - **Correct** them if caused by obvious error in recording

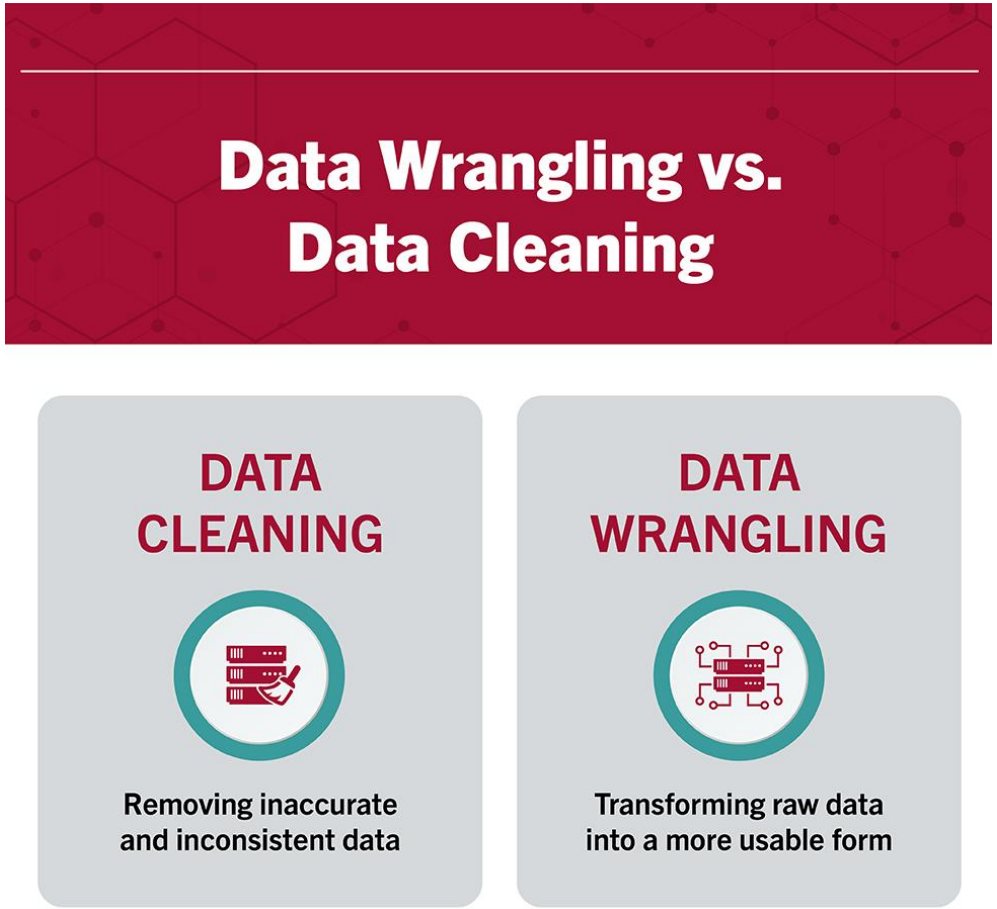
Data Wrangling vs. Data Cleaning

- Data Wrangling
  - The **overall process** of transforming raw data into a usable, analysis-ready format
  - Includes multiple steps: *discovering, structuring, cleaning, enriching, validating, publishing*
- Data Cleaning
  - A **specific step** within data wrangling
  - Focuses on *detecting and correcting errors, inconsistencies, and missing values*

In short:

All data cleaning is part of data wrangling,

but data wrangling goes beyond cleaning.



Data Wrangling – Enriching

Goal

- Add **new**, **relevant** information to existing datasets to enhance their value and insight *(Also known as data augmentation.)*

What It Involves

- Identify what additional information is needed and **where to source it**
- Integrate and clean new data using the same wrangling steps (discovery, structuring, cleaning)

Key Approaches

- **Subsetting:** extract only fields or records relevant to analysis
- **Supplementing:** merge external data
  - *One-to-one merge* → each record matched once
  - *One-to-many merge* → one record matched with multiple entries
- **Enhancing:** transform or recode existing fields (e.g., format, structure, or value adjustments)

Data Wrangling – Validating and Publishing

Goal

- Verify that the *cleaned* and *enriched* data are accurate, consistent, and ready for use

Common Validation Checks

- Data type validation: confirm correct variable types
- Range / format checks: ensure values fall within valid limits or match expected formats
- Consistency checks: verify logical agreement between related fields
- Uniqueness checks: confirm IDs or keys are not duplicated
- Cross-field validation: check relationships (e.g., age vs. birthdate)
- Statistical analysis: detect outliers or anomalies using summary stats or visualizations

After Validation

- Publish or export data for analytics, visualization, or machine learning
- Load into databases or data warehouses for broader use

Summary

- Today
  - What is data wrangling?
    - Transform raw, messy data into clean, reliable, analysis-ready form
  - Why it matters?
    - High-quality data → accurate insights, better decisions, trustworthy models
  - How it works?
    - Six key steps: discovering, structuring, cleaning, enriching, validating, publishing
  - Data Cleaning
    - Missing values, normalization & scaling, outliners
- Looking Ahead
  - Oct 16 (Thu): Review for midterm exam
  - Oct 18 (Sat): Assignment 2 due
  - Oct 21 (Tue): Midterm exam, 7:00 – 8:00 PM?
  - Oct 23 (Thu): Data cleaning with Python